

MATH 350-2 Advanced Calculus

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Outline

- 1 Real Analysis Lecture 4
 - Functions
 - More with cardinality

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 - Functions
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Challenge

- index set $I = (1, 2)$
- family of sets $\{A_i : i \in I\}$
- $A_i = [0, i]$

Problem

Determine $\bigcup_{i \in I} A_i$.

Challenge

- index set $I = \mathbb{N}$
- family of sets $\{A_i : i \in I\}$
- $A_i = [0, 1/i)$

Problem

Determine $\bigcap_{i \in I} A_i$.

Relations

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The set

$$\text{img}(f) = \{f(a) : a \in A\}$$

is called the **range** or **image** of f

Challenge!

Problem

Is the relation on $\mathbb{R}$ defined by

$$\{(x, y) : x^2 + y^2 = 1\}$$

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If it satisfies both properties, it is called **bijective**.

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If it is, we call it the **inverse** of f .

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$$(g \circ f)(x) = g(f(x)).$$

Cantor's Paradise

Definition

Two sets A and B have the same cardinality if there exists a bijection $f : A \rightarrow B$.

NOTATION: $|A| = |B|$

- $|\mathbb{Z}| = |\mathbb{Z}_+|$
- $|\mathbb{Q}| = |\mathbb{Z}_+|$

Cantor's Paradise

Definition

We say the cardinality of A is less than or equal to B if there exists an injection $f : A \rightarrow B$. We say the cardinality of A is greater than or equal to B if there exists a surjection $f : A \rightarrow B$.

NOTATION: $|A| \leq |B|$ and $|A| \geq |B|$

- $|\mathbb{Z}| \leq |\mathbb{R}|$
- $|\mathbb{Z}| \geq |\mathbb{N}|$

Cantor's Paradise

Theorem (Cantor-Schroeder-Bernstein Theorem)

The following are equivalent

- (i) $|A| \leq |B|$ and $|B| \leq |A|$
- (ii) $|A| \geq |B|$ and $|B| \geq |A|$
- (iii) $|A| \leq |B|$ and $|A| \geq |B|$
- (iv) $|A| = |B|$

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Hint: consider $f : \mathbb{Z}_+ \times \mathbb{Z}_+ \rightarrow \mathbb{Z}_+$, $f(m, n) = 2^m 3^n$

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Hint: consider $f : \mathbb{Z}_+ \rightarrow \mathbb{Z}$

$$f(n) = \begin{cases} n/2 & n \text{ is even} \\ -(n-1)/2 & n \text{ is odd} \end{cases}$$

Finite and infinite

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Sets with finite cardinality:

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Cantor's discovery: there are multiple sizes of infinity!

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$$\mathbb{R} \text{ has larger cardinality than } \mathbb{Z}_+.$$

Cantor diagonalization

Suppose there were a bijection

$$f : \mathbb{Z}_+ \rightarrow (0, 1)$$

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$$f(1) = 0.a_{11}a_{12}a_{13}a_{14}a_{15}\dots$$

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$$f(3) = 0.a_{31}a_{32}a_{33}a_{34}a_{35}\dots$$

$$f(4) = 0.a_{41}a_{42}a_{43}a_{44}a_{45}\dots$$

$$f(5) = 0.a_{51}a_{52}a_{53}a_{54}a_{55}\dots$$

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Cantor diagonalization

Now define $b_1, b_2, b_3 \cdots \in \{0, 9\}$ by

$$b_j = \begin{cases} 1, & a_{jj} = 0 \\ 0, & a_{jj} \neq 0 \end{cases}$$

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Theorem (Cantor's Theorem)

$\mathbb{R}$ is uncountable

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Ultimate Cantor diagonalization

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Proof.

Suppose $f : A \rightarrow \mathcal{P}(A)$ is surjective.
Consider the set

$$S = \{a \in A : a \notin f(a)\}.$$



Challenge

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Problem

Consider the statement $x \in S$. What can you conclude?

Challenge

Problem

Is there a set with cardinality larger than $\mathbb{R}$?

Challenge

Problem

Show that the cardinality of the line segment $(0, 1)$ and the square $(0, 1) \times (0, 1)$ is the same.

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$f(j, k) = a_{i_j, k}$. Surjection!



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Hint: use the previous theorem!